

Evolutionary Optimization of Activation Function Distributions

A Genetic Algorithm Approach

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Overview

Setting

Consider a neural network with one or more hidden layers.

Notation:

- $v = \{v_1, v_2, \dots, v_n\}$ — raw output of a hidden layer
- $f = \{f_1, f_2, \dots, f_n\}$ — activation functions applied per neuron
- $f_i \in \mathcal{F}$ for $i = 1, \dots, n$
- $\mathcal{F}$ — the **function pool** (set of candidate activation functions)

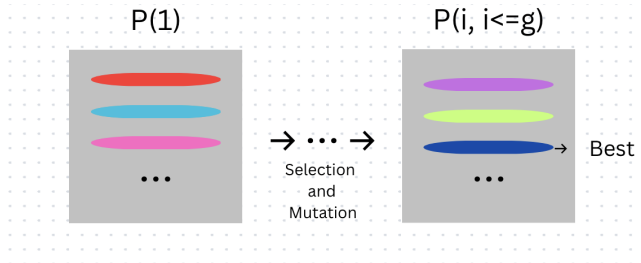
Goal

Optimize the *distribution* of activation functions across hidden layer neurons using a genetic algorithm.

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Algorithm Overview - In a Nutshell



Algorithm Step by Step

- Populate initial generation P^0 with M individuals
- Populate next generation with selection operator and evaluate each individual f with fitness function (validation loss)
- Mutate individuals in next generation
- Return best individual f_{best} when generation reached its maximal or f_{best} satisfies criteria

Algorithm Overview

```
1: Initialize  $P^0$  with random functions from  $\mathcal{F}$ 
2: Initialize  $f_{best}$ 
3: for  $t = 0$  to  $g$  do
4:   for  $i = 1$  to  $M$  do
5:     Evaluate fitness of  $P_i^t$  by training NN with  $P_i^t$ 
6:     if  $\text{fitness}(P_i^t) > \text{fitness}(f_{best})$  then
7:        $f_{best} \leftarrow P_i^t$ 
8:     end if
9:     if  $f_{best}$  satisfies criteria then
10:      return  $f_{best}$ 
11:    end if
12:  end for
13:  Select new population  $P^{t+1}$  from  $P^t$ 
14:  Mutate  $P^{t+1}$ 
15: end for
16: return  $f_{best}$ 
```

Multi-layer Extension

Single Hidden Layer (Default)

Each individual in the population is a vector of activation functions of length n (number of neurons).

$N > 1$ Hidden Layers

Two possible representations:

- **Concatenation:** stack the activation function vectors of all layers into one individual
- **Alternative encoding:** to be determined per experiment

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Selection

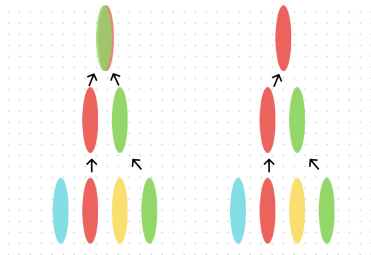
Strategy: Elitism + Tournament Selection (initial $k = 2$)

Algorithm 1 Selection(k)

```
1: Select  $f_{best} \in P^t$ ; add to  $P^{t+1}$ 
2: while  $P^{t+1}$  is not full do
3:   Sample  $k$  individuals  $\rightarrow$  pick best as  $f_A$ 
4:   Sample  $k$  individuals  $\rightarrow$  pick best as  $f_B$ 
5:   Execute Crossover( $f_A, f_B$ ) with probability  $p_c$ 
6:   if Crossover executed then
7:     Add child  $f_C$  to  $P^{t+1}$ 
8:   else
9:     Add  $\arg \max_{\{f_A, f_B\}} \text{fitness}(f)$  to  $P^{t+1}$ 
10:  end if
11: end while
```

- Tournament size: $k = 2$ (4 individuals compared per slot)
- Elitism ensures the best solution is never lost

Tournament Selection In a Nutshell



Example with $k = 2$

- Select two individual pairs from P^t
- Pick the best one from each pair (parents)
- If Crossover (left): Make new individual with crossover from parents
- Else (right): Pick the best one from parents
- Add new individual in P^{t+1}

Crossover

Strategy: Uniform Crossover — two parents produce one child

Algorithm 2 Crossover(f_A, f_B)

- 1: Initialize f_C
 - 2: **for** each (f_{A_i}, f_{B_i}) in f_A, f_B **do**
 - 3: $f_{C_i} = f_{A_i}$ with probability 0.5
 - 4: Otherwise, $f_{C_i} = f_{B_i}$
 - 5: **end for**
 - 6: **return** f_C
-

Parameters

- Initial crossover probability: $p_c = 0.8$
- Will remain **high** across experiments

Mutation

Strategy: Random replacement from the function pool

Algorithm 3 Mutation(f, l)

- 1: Select l elements from f uniformly at random
 - 2: **for** each selected element f_i **do**
 - 3: Replace f_i with a randomly drawn $f_s \in \mathcal{F} \setminus \{f_i\}$
 - 4: **end for**
-

Parameters

- Initial mutation probability: $p_m = 0.01$ (kept very low)
- Number of mutated elements: $l \approx 20\%$ of individual length

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Summary of Parameters

Parameter	Initial Value	Notes
Population size M	—	To be set per experiment
Max generations g	—	To be set per experiment
Tournament size k	2	4 individuals compared
Crossover prob. p_c	0.8	Remains high
Mutation prob. p_m	0.01	Remains very low
Mutated elements l	$\approx 20\%$ of length	Per individual

Key Design Choices

- Elitism preserves the best-found solution at all times
- Uniform crossover provides unbiased gene mixing
- Low mutation rate maintains population stability